

ReCodeAgent paper Tables 1 and 2 with CodeWeaver results

Paper values: arXiv:2604.07341 and the official results.xlsx (MD5 a404779f2dcd7ac44d43bf72f4e88b98). CodeWeaver values: variant='full', repetition=0. P-Tool is the paper's comparison tool; P-RCA is the paper's ReCodeAgent result; CW is independently measured CodeWeaver. A trailing * marks a partial measured sum/mean where one or more constituent CodeWeaver cells were unavailable. '-' means unavailable or not applicable and is never converted to zero.

Table 1 - Effectiveness

Effectiveness of ReCodeAgent in repository-level code translation and validation in terms of test and function validation. Tuple components from the paper are expanded into P-Tool and P-RCA columns; CodeWeaver is placed immediately beside the corresponding paper metric. E=expected, TE=executed, TP=passing, TF=failing, C=coverage before generated tests, C+=coverage after generated tests, T=total, S=success, F=fail.

Tool (PL)	Project	LoC	CS % P-Tool	CS % P-RCA	CS % CW	Validated developer tests E/TE/TP/TF P-Tool	Validated developer tests E/TE/TP/TF P-RCA	Validated developer tests E/TE/TP/TF CW	Translated developer tests TE/TP/TF P-RCA	Translated developer tests TE/TP/TF CW	Generated tests TE/TP/TF P-RCA	Generated tests TE/TP/TF CW	Coverage C/C+ % P-RCA	Coverage C/C+ % CW	Function validation T/S/F P-Tool	Function validation T/S/F P-RCA	Function validation E/TE/S/F CW
Oxidizer (Go->Rust)	checkdigit	428	100	100	100	36/36/33/3	36/36/36/0	36/36/36/0	36/36/0	36/36/0	71/71/0	0/0/0	79.7/94.7	93.0/93.0	29/21/8	29/29/0	29/12/12/0
Oxidizer (Go->Rust)	go-edlib	639	100	100	100	36/36/19/17	36/36/36/0	36/7/7/0	36/36/0	48/48/0	3/3/0	0/0/0	94.7/94.9	75.6/75.6	24/18/6	24/24/0	24/7/7/0
Oxidizer (Go->Rust)	histogram	314	100	100	100	2/2/2/0	2/2/2/0	2/3/3/0	2/2/0	4/4/0	66/66/0	2/2/0	38.0/90.5	48.6/48.6	19/12/7	19/19/0	19/-/-
Oxidizer (Go->Rust)	nameparts	413	100	100	100	26/26/23/3	26/26/26/0	26/-/-/-	26/26/0	32/32/0	22/22/0	5/5/0	96.8/96.8	-	15/9/6	15/14/1	15/-/-/-
Oxidizer (Go->Rust)	stats	1,241	100	100	100	121/121/71/50	121/121/121/0	121/9/9/0	121/121/0	107/107/0	320/320/0	0/0/0	43.3/79.6	3.1/3.1	52/38/14	52/52/0	52/-/-/-
Oxidizer (Go->Rust)	textrank	1,132	100	100	100	8/8/6/2	8/8/8/0	8/3/3/0	8/8/0	12/11/1	127/127/0	9/9/0	72.6/98.7	36.1/49.6	52/40/12	52/52/0	52/-/-/-
Oxidizer (Go->Rust)	Total	4,167	100	100	100	229/229/154/75	229/229/229/0	229/58*/58*/0*	229/229/0	239/238/1	609/609/0	16/16/0	70.8/92.5	51.3*/54.0*	191/138/53	191/190/1	191/19*/19*/0*
AlphaTrans (Java->Python)	cli	37,841	100	100	100	381/66/35/31	381/381/360/21	381/381/201/180	381/381/0	-	257/257/0	18/18/0	96.7/97.3	87.6/89.9	257/196/61	257/241/16	257/-/-/-
AlphaTrans (Java->Python)	csv	33,072	100	100	100	298/147/3/144	298/298/241/57	298/-/-/-	298/298/0	-	192/190/2	151/151/0	84.4/85.8	29.9/57.5	213/74/139	213/211/2	213/-/-/-
AlphaTrans (Java->Python)	fileupload	3,567	100	100	100	39/39/36/3	39/39/39/0	39/9/6/3	39/39/0	-	208/208/0	160/160/0	38.7/71.6	35.3/79.0	25/19/6	25/25/0	25/-/-/-
AlphaTrans (Java->Python)	validator	41,605	100	100	100	463/359/114/245	463/463/438/25	463/20/0/20	463/435/28	-	132/131/1	13/13/0	65.0/73.0	18.5/19.7	409/217/192	409/397/12	409/-/-/-
AlphaTrans (Java->Python)	Total	116,085	100	100	100	1,181/611/188/423	1,181/1,181/1,078/103	1,181/410*/207*/203*	1,181/1,153/28	-	789/786/3	342/342/0	71.2/81.9	42.8/61.5	904/506/398	904/874/30	904/-/-/-
SKEL (Python->JavaScript)	bst	123	100	100	100	11/11/11/0	11/11/11/0	11/11/0/11	11/11/0	-	6/6/0	0/0/0	89.7/99.0	41.3/41.3	21/21/0	21/21/0	21/-/-/-
SKEL (Python->JavaScript)	colorsys	120	100	100	100	2/2/2/0	2/2/2/0	2/2/2/0	2/2/0	-	46/46/0	0/0/0	87.0/91.3	80/80	9/9/0	9/9/0	9/-/-/-
SKEL (Python->JavaScript)	heapq	189	100	100	100	8/8/7/1	8/8/8/0	8/8/8/0	8/8/0	-	11/11/0	0/0/0	91.6/91.6	85.6/85.6	24/23/1	24/24/0	24/-/-/-
SKEL (Python->JavaScript)	html	684	100	100	100	7/7/6/1	7/7/7/0	7/-/-/-	7/7/0	-	13/13/0	2/2/0	77.1/86.1	93.3/93.3	42/39/3	42/42/0	42/-/-/-
SKEL (Python->JavaScript)	mathgen	735	100	100	100	5/5/4/1	5/5/5/0	5/5/0/5	5/5/0	-	11/11/0	0/0/0	96.4/98.6	39.7/39.7	82/79/3	82/82/0	82/-/-/-
SKEL (Python->JavaScript)	rbt	366	100	100	100	10/10/10/0	10/10/10/0	10/10/6/4	10/10/0	-	5/5/0	0/0/0	87.0/88.3	81.3/81.3	27/27/0	27/27/0	27/-/-/-
SKEL (Python->JavaScript)	strsim	654	100	100	100	19/19/19/0	19/19/19/0	19/19/0/19	19/19/0	-	64/64/0	0/0/0	88.8/94.1	25.9/25.9	50/50/0	50/50/0	50/-/-/-
SKEL (Python->JavaScript)	toml	1,206	100	100	100	12/12/10/2	12/12/12/0	12/12/2/10	12/12/0	-	150/150/0	2/2/0	72.6/83.2	31.9/48.0	47/43/4	47/47/0	47/-/-/-
SKEL (Python->JavaScript)	Total	4,077	100	100	100	74/74/69/5	74/74/74/0	74/67*/18*/49*	74/74/0	-	306/306/0	4/4/0	86.3/91.5	59.9/61.9	302/291/11	302/302/0	302/-/-/-
SWE-agent (C->Rust)	Crust- α	22,961	40	40	100*	166/153/130/23	166/166/146/20	166/688*/688*/0*	-	840*/840*/0*	493/493/0	654/653/1	68.2/75.7	69.0*/79.9*	673/-/-	673/-/-	-
SWE-agent (C->Rust)	Crust- β	66,704	0	49	100	321/0/0/0	321/320/295/25	321/594*/592*/2*	-	724*/722*/2*	1,118/1,114/4	373/372/1	57.9/75.2	62.1*/70.0*	1,900/-/-	1,900/-/-	-
SWE-agent (C->Rust)	Crust- σ	3,894	1	0	-	1/1/1/0	1/0/0/0	1/-/-/-	-	-	53/51/2	0/0/0	4.3/67.4	-	41/-/-	41/-/-	-
SWE-agent (C->Rust)	Crust- γ	15,169	0	0	100*	135/0/0/0	135/0/0/0	135/150*/148*/2*	-	194*/192*/2*	274/270/4	60/60/0	27.0/44.7	41.5*/45.4*	572/-/-	572/-/-	-
SWE-agent (C->Rust)	Total	108,728	41	89	100*	623/154/131/23	623/486/441/45	623/1,432*/1,428*/4*	-	1,758*/1,754*/4*	1,938/1,928/10	1,087/1,085/2	39.3/65.7	57.5*/65.1*	3,186/-/-	3,186/-/-	-
Total	Total	233,057	96.9	99.4	100*	2,107/1,068/542/526	2,107/1,970/1,822/148	2,107/1,967*/1,711*/256*	1,484/1,456/28	1,997*/1,992*/5*	3,642/3,629/13	1,449/1,447/2	70.8/85.4	54.0*/60.3*	4,583/935/462	4,583/1,366/31	1,397/19*/19*/0*

Table 2 - Test Translation

Comparison between ReCodeAgent translated tests and original source-language tests, with the same official AST/runtime protocol applied to CodeWeaver. Paper subtotal means follow the workbook's SUBTOTAL(AVERAGE) formulas. M/N=matching/non-matching, T/M=total/matching, S/T=source/target.

Tool (PL)	Project	Tests Paper	Tests CW	Translated/Not Paper	Translated/Not CW	Assertion count M/N Paper	Assertion count M/N CW	assertEqual output T/M Paper	assertEqual output T/M CW	Type match Equal % Paper	Type match Equal % CW	Type match True % Paper	Type match True % CW	Type match False % Paper	Type match False % CW	Type match Other % Paper	Type match Other % CW	Avg cosine Paper	Avg cosine CW	Avg LoC S/T Paper	Avg LoC S/T CW	Avg calls S/T Paper	Avg calls S/T CW
Oxidizer (Go->Rust)	checkdigit	36	36	36/0	24/12	36/0	12/24	45/45	0/0	100	100	-	-	-	-	-	-	0.94	0.85	24.42/24.58	29.96/12.83	7.14/5.33	6.71/8.96
Oxidizer (Go->Rust)	go-edlib	36	36	36/0	36/0	36/0	23/13	45/45	0/0	84.44	97.78	-	-	-	-	-	-	0.92	0.86	30.78/51	30.78/45.25	7.61/9.50	7.61/16.28
Oxidizer (Go->Rust)	histogram	2	2	2/0	2/0	2/0	2/0	11/11	0/0	100	100	-	-	-	-	-	-	0.96	0.86	23.50/16.50	23.50/19	24/16.50	24/16.50
Oxidizer (Go->Rust)	nameparts	26	26	26/0	24/2	26/0	20/6	51/51	0/0	100	92.16	-	-	-	-	-	-	0.94	0.88	11.96/6.31	11.96/5.58	6.15/3.81	6.12/2.67
Oxidizer (Go->Rust)	stats	121	121	121/0	104/17	121/0	60/61	150/150	0/0	91.15	88.66	-	-	-	-	-	-	0.85	0.85	16.82/9.44	15.28/10.97	8.12/6	8.19/8.84
Oxidizer (Go->Rust)	textrank	8	8	8/0	7/1	8/0	1/7	12/12	0/0	100	-	-	-	-	-	-	-	0.91	0.87	13.75/15.25	10.43/9.57	10.88/11.88	10.43/7.29
Oxidizer (Go->Rust)	Total	229	229	229/0	197/32	229/0	118/111	314/314	0/0	95.93	95.72	-	-	-	-	-	-	0.92	0.86	20.21/20.51	20.32/17.20	10.65/8.84	10.51/10.09
AlphaTrans (Java->Python)	cli	381	381	381/0	267/114	381/0	187/194	452/452	220/212	99.61	75.65	99.68	75.15	98.37	32.20	97.87	26.74	0.90	0.67	12.41/11.59	12.72/9.91	13.30/12.63	12.69/8.18
AlphaTrans (Java->Python)	csv	298	298	298/0	128/170	292/6	100/198	207/207	99/88	100	78.10	81.82	26.92	84.31	7.87	92.86	18.52	0.90	0.69	11.84/8.94	13.94/12.03	12.20/10.46	14.68/8.76
AlphaTrans (Java->Python)	fileupload	39	39	39/0	39/0	39/0	35/4	37/37	30/30	100	83.78	100	66.67	80	20	100	50	0.87	0.66	6.74/7.38	6.74/7.49	5.87/6.36	5.87/4.92
AlphaTrans (Java->Python)	validator	463	463	463/0	239/224	458/5	154/309	374/374	131/125	98.52	63.66	99.28	57.81	99.49	44.04	90.72	12.87	0.89	0.73	17.69/17.23	20.78/10.74	18.78/18.30	23.13/8.77
AlphaTrans (Java->Python)	Total	1,181	1,181	1,181/0	673/508	1,170/11	476/705	1,070/1,070	480/455	99.53	75.30	95.19	56.64	90.54	26.03	95.36	27.03	0.89	0.69	12.17/11.29	13.54/10.04	12.54/11.94	14.09/7.66
SKEL (Python->JavaScript)	bst	11	11	11/0	0/11	11/0	0/11	74/74	0/0	100	-	-	-	-	-	-	-	0.91	-	22.27/22.64	-	5.73/12	-
SKEL (Python->JavaScript)	colorsys	2	2	2/0	2/0	2/0	0/2	39/39	0/0	100	-	-	-	-	-	-	-	0.93	0.92	67/65	67/66.50	45/44	45/44
SKEL (Python->JavaScript)	heapq	8	8	8/0	8/0	8/0	6/2	29/29	0/0	100	-	-	-	-	-	-	-	0.90	0.91	13.12/13.38	13.12/11.12	11.25/10.25	11.25/10.25
SKEL (Python->JavaScript)	html	7	7	7/0	7/0	7/0	6/1	19/19	0/0	100	-	-	-	-	-	-	-	0.92	0.92	24.71/22.71	24.71/24.57	20.29/19.29	20.29/19.14
SKEL (Python->JavaScript)	mathgen	5	5	5/0	5/0	5/0	0/5	163/163	0/0	100	-	-	-	-	-	-	-	0.93	0.93	47/51	47/52	49/42	49/43
SKEL (Python->JavaScript)	rbt	10	10	10/0	10/0	10/0	2/8	16/16	0/0	100	-	-	-	-	-	-	-	0.91	0.91	17.90/19.60	17.90/20.20	20.90/17.10	20.90/20
SKEL (Python->JavaScript)	strsim	19	19	19/0	19/0	19/0	0/19	150/150	0/0	100	-	-	-	-	-	-	-	0.93	0.92	18.74/17.21	18.74/15.79	22.84/21.84	22.84/21.84
SKEL (Python->JavaScript)	toml	12	12	12/0	12/0	12/0	0/12	17/17	0/0	100	-	-	-	-	-	-	-	0.90	0.86	8.67/8.75	8.67/13	6.08/7	6.08/11.83
SKEL (Python->JavaScript)	Total	74	74	74/0	63/11	74/0	14/60	507/507	0/0	100	-	-	-	-	-	-	-	0.92	0.91	27.43/27.54	28.16/29.03	22.64/21.68	25.05/24.29
Total	Total	1,484	1,484	1,484/0	933/551	1,473/11	608/876	1,891/1,891	480/455	98.54	86.64	95.19	56.64	90.54	26.03	95.36	27.03	0.91	0.84	21.63/21.58	21.95/20.39	16.40/15.24	17.34/15.37